Week 1

1. A dataset was highly useful in 2024, but by 2026 the same dataset becomes irrelevant because the information has lost significance.

Which "V" best describes this situation?

A. Validity
B. Veracity
C. Volatility
D. Venue

Answer: C

Explanation

A. Validity (Incorrect): Validity refers to whether the data is correct, accurate, and suitable for its intended purpose.

B. Veracity (Incorrect): Veracity relates to the trustworthiness, quality, and reliability of the data.

C. Volatility (Correct): Volatility refers to how long data remains useful and relevant before it loses value or becomes outdated.

D. Venue (Incorrect): Venue refers to the source or location from which data is collected, not its lifespan or relevance.

2. **Which of the following statements about Oozie is CORRECT?**

A) Oozie is a distributed NoSQL database.
B) Oozie is a workflow scheduler system for Hadoop jobs.
C) Oozie is used to store data in HDFS.
D) Oozie is a machine learning library for Spark.

Answer: B)

Explanation

A. Oozie is a distributed NoSQL database (Incorrect): NoSQL databases include systems like Cassandra, HBase, and MongoDB; Oozie is not a database.

B. Oozie is a workflow scheduler system for Hadoop jobs (Correct): Oozie is used to define, schedule, and manage workflows consisting of Hadoop ecosystem jobs such as MapReduce, Pig, Hive, and Sqoop.

C. Oozie is used to store data in HDFS (Incorrect): HDFS stores data, while Oozie manages and schedules workflows that may operate on that data.

D. Oozie is a machine learning library for Spark (Incorrect): Spark's machine learning library is MLlib; Oozie is a workflow orchestration tool.

3. **Oozie can directly coordinate which of the following Hadoop ecosystem components?**

A) Hive, Pig, Sqoop, and MapReduce
B) HDFS, NameNode, DataNode, and YARN only
C) Cassandra, MongoDB, and Redis only
D) Spark MLlib and GraphX only

Answer: A

Explanation

A. Hive, Pig, Sqoop, and MapReduce (Correct): Oozie can schedule and coordinate workflows involving Hadoop ecosystem tools such as Hive, Pig, Sqoop, MapReduce, and others.

B. HDFS, NameNode, DataNode, and YARN only (Incorrect): These are Hadoop storage and resource management components; Oozie coordinates jobs running on the ecosystem rather than directly managing these components.

C. Cassandra, MongoDB, and Redis only (Incorrect): These are external NoSQL databases and are not the primary Hadoop ecosystem components coordinated by Oozie.

D. Spark MLlib and GraphX only (Incorrect): MLlib and GraphX are Spark libraries; Oozie's core purpose is broader workflow orchestration and is not limited to these libraries.

4. **What is the primary purpose of Apache Flume in the Hadoop ecosystem?**

A) Querying data stored in HDFS
B) Collecting and transferring large amounts of log data

C) Performing machine learning tasks
D) Managing cluster resources

Answer: B

Explanation

A. Querying data stored in HDFS (Incorrect): Querying HDFS data is typically performed using tools like Hive, Impala, or Spark SQL, not Flume.

B. Collecting and transferring large amounts of log data (Correct): Apache Flume is a distributed service designed to efficiently collect, aggregate, and move large volumes of log and event data into Hadoop systems such as HDFS.

C. Performing machine learning tasks (Incorrect): Machine learning is handled by frameworks such as Spark MLlib or Mahout, not Flume.

D. Managing cluster resources (Incorrect): Resource management in Hadoop is handled by YARN, while Flume focuses on data ingestion.

5. **What is the primary role of Apache ZooKeeper in a Hadoop ecosystem?**

A) Distributed data storage
B) Workflow scheduling
C) Distributed coordination and synchronization
D) SQL query processing

Answer: C

Explanation

A. Distributed data storage (Incorrect): ZooKeeper stores small amounts of coordination metadata, not large-scale application data like HDFS or HBase.

B. Workflow scheduling (Incorrect): Workflow scheduling is primarily handled by tools such as Oozie, not ZooKeeper.

C. Distributed coordination and synchronization (Correct): Apache ZooKeeper provides centralized coordination services such as configuration management, synchronization, naming, and leader election for distributed systems.

D. SQL query processing (Incorrect): SQL query processing is performed by tools like Hive or Spark SQL, whereas ZooKeeper focuses on coordination services.

6. **Hive primarily operates on data stored in:**

A) MongoDB
B) Cassandra
C) HDFS
D) Redis

Answer: C

Explanation

A. MongoDB (Incorrect): MongoDB is a NoSQL document database and is not Hive's primary storage layer.

B. Cassandra (Incorrect): Cassandra is a distributed NoSQL database; Hive is mainly designed to work with data stored in HDFS.

C. HDFS (Correct): Apache Hive is a data warehouse tool that primarily stores and processes structured data residing in the Hadoop Distributed File System (HDFS).

D. Redis (Incorrect): Redis is an in-memory key-value store and is not the default storage system used by Hive.

7. **Which of the following best describes Apache Hive?**

A) A distributed coordination service
B) A workflow scheduler
C) A data warehouse system built on Hadoop
D) A NoSQL database

Answer: C

Explanation

A. A distributed coordination service (Incorrect): Distributed coordination is the role of Apache ZooKeeper, not Hive.

B. A workflow scheduler (Incorrect): Workflow scheduling and job orchestration are handled by Apache Oozie, not Hive.

C. A data warehouse system built on Hadoop (Correct): Apache Hive is a **data warehouse infrastructure** that provides SQL-like querying (HiveQL) for analyzing large datasets stored in Hadoop, primarily in HDFS.

D. A NoSQL database (Incorrect): Hive is not a NoSQL database; it is a data warehousing and query system built on top of Hadoop storage.

8. Which Hadoop module stores metadata?

A) DataNode
B) TaskTracker
C) NameNode
D) Reducer

Answer: C

Explanation

A. DataNode (Incorrect): DataNodes store the actual data blocks, not the filesystem metadata.

B. TaskTracker (Incorrect): TaskTracker (in Hadoop 1.x) executes MapReduce tasks and does not manage HDFS metadata.

C. NameNode (Correct): The NameNode stores HDFS metadata, including file names, directory structure, permissions, and the mapping of files to data blocks.

D. Reducer (Incorrect): A Reducer processes intermediate key-value pairs in MapReduce and has no role in storing HDFS metadata.

9. Which pair is correctly matched?

A) Hive – Stream Processing
B) Kafka – Distributed Commit Log
C) Sqoop – Graph Analytics
D) GraphX – ETL Tool

Answer: B

Explanation

A. Hive – Stream Processing (Incorrect): Hive is a data warehouse and SQL-like query tool for batch processing, not a stream processing system.

B. Kafka – Distributed Commit Log (Correct): Apache Kafka is a distributed event streaming platform built around a distributed commit log architecture for storing and processing streams of records.

C. Sqoop – Graph Analytics (Incorrect): Sqoop is used for transferring data between relational databases and Hadoop, not for graph analytics.

D. GraphX – ETL Tool (Incorrect): GraphX is Spark's graph processing and analytics library, not an ETL (Extract, Transform, Load) tool.

10. **Which of the following is the PRIMARY function of Apache ZooKeeper in a distributed Hadoop environment?**

A) Storing large-scale data across cluster nodes
B) Managing distributed coordination, synchronization, and configuration services
C) Executing SQL-like queries on HDFS data
D) Scheduling Hadoop workflows and jobs

Correct Answer: B

Explanation

A. Storing large-scale data across cluster nodes (Incorrect): Large-scale data storage is handled by HDFS or HBase; ZooKeeper stores only small amounts of coordination metadata.

B. Managing distributed coordination, synchronization, and configuration services (Correct): ZooKeeper provides services such as leader election, configuration management, naming, synchronization, and coordination among distributed applications.

C. Executing SQL-like queries on HDFS data (Incorrect): SQL-like queries on Hadoop data are performed using tools such as Hive or Spark SQL, not ZooKeeper.

D. Scheduling Hadoop workflows and jobs (Incorrect): Workflow scheduling is the responsibility of Oozie, whereas ZooKeeper focuses on distributed coordination services.